

Computer Vision and Image Processing Project (2025/26)

Deep Learning for Traffic Sign Recognition using the GTSRB Dataset

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Abstract—This report presents a deep learning project focused on traffic sign recognition using the German Traffic Sign Recognition Benchmark (GTSRB) dataset. The main objective was to study the impact of different data augmentation techniques, image preprocessing strategies and ensemble learning techniques on the performance of convolutional neural networks (CNNs). Several augmentation configurations were evaluated, including geometric transformations, color-based augmentations, combined augmentations, massive dataset expansion and MixUp, followed by the construction of ensembles using Soft Voting and Hard Voting methods. Experimental results demonstrate that data augmentation significantly improves model generalization and robustness, while ensemble learning enhances classification performance by leveraging the diversity of independently trained models and reducing prediction variance. The best ensemble configuration achieved an accuracy of up to 98.80%, confirming that combining complementary models leads to more accurate and stable predictions on the GTSRB benchmark.

Index Terms—Deep Learning, Computer Vision, Traffic Sign Recognition, GTSRB, Convolutional Neural Networks, Data Augmentation, Image Processing, Ensemble Learning

I. INTRODUCTION

Traffic sign recognition is an important task in computer vision and autonomous driving systems. Modern Advanced Driver Assistance Systems (ADAS) rely heavily on robust visual recognition models capable of detecting and classifying traffic signs under different environmental conditions such as illumination changes, rotations, occlusions, and motion blur.

The objective of this project is to explore deep learning techniques applied to the GTSRB dataset in order to achieve the highest possible classification accuracy. According to published benchmarks, the best reported accuracy on the dataset is approximately 99.82%.

In the first phase, several image preprocessing and data augmentation techniques are evaluated. The goal is to understand how different augmentation strategies influence model generalization and robustness.

In the second phase, ensemble learning techniques are explored. Multiple models trained during Phase 1 are combined using voting strategies in order to reduce prediction variance and improve overall classification performance.

II. DATASET

The German Traffic Sign Recognition Benchmark (GTSRB) is a well-known benchmark dataset for traffic sign classification.

The dataset contains:

- 43 traffic sign classes;
- More than 50,000 images;
- Images captured under real-world conditions;
- Significant variability in illumination, perspective, scale, and occlusion.

The dataset was divided into:

- Training set;
- Validation set;
- Test set.

A. Class Distribution

One important characteristic of the dataset is the imbalance between classes. Some traffic signs contain considerably more samples than others, which can introduce bias during training.

The initial dataset analysis revealed:

- Strong variability in the number of samples per class;
- Variability in image resolution;
- Presence of noisy samples and illumination changes.

III. METHODOLOGY

A. Framework and Libraries

The project was implemented using:

- Python;
- PyTorch;
- Torchvision;
- NumPy;
- Matplotlib;
- Pandas.

B. Image Preprocessing

All images were resized to a fixed spatial resolution before being provided to the neural networks. Resizing ensures consistent tensor dimensions and allows efficient mini-batch processing during training.

The preprocessing pipeline included:

- Image resizing;
- Conversion to tensor format;
- Floating-point normalization;
- Standardization when required by pretrained models.

Example preprocessing pipeline:

```
transform_base = v2.Compose([
    v2.Resize((IMG_SIZE, IMG_SIZE)),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True)
])
```

For transfer-learning experiments using VGG16, ImageNet normalization statistics were applied to match the distribution of the pretrained weights.

IV. MODEL ARCHITECTURE

Three different convolutional architectures were evaluated.

A. Custom CNN (Conv_II)

A custom convolutional neural network was developed specifically for the GTSRB dataset. The architecture includes:

- Multiple convolutional layers;
- Batch normalization;
- ReLU activations;
- Max-pooling operations;
- Fully connected classification layers;
- Dropout regularization.

This architecture was designed to balance computational efficiency and classification accuracy.

B. VGG16

The second architecture evaluated was VGG16. In this configuration, the network was initialized without pretrained weights and trained directly on the GTSRB dataset. The original classifier was replaced with a custom classification head composed of a fully connected layer with 256 neurons, a ReLU activation function, a dropout layer with a rate of 0.5, and a final output layer corresponding to the 43 traffic sign classes.

For this architecture, images were resized to a dedicated input resolution and normalized before being fed to the network. The objective was to evaluate the performance of a deeper architecture when trained exclusively on the available traffic sign data.

C. VGG16 with Transfer Learning

A second VGG16-based approach was implemented using transfer learning. In this configuration, the network was initialized with pretrained weights, and all feature extraction layers were frozen during training. The original classifier was replaced by the same custom classification head used in the previous experiment, consisting of a fully connected layer with 256 neurons, ReLU activation, dropout regularization, and a final classification layer with 43 outputs.

Only the classifier layers were updated during training, while the pretrained feature extraction layers remained unchanged. This approach was evaluated to determine whether reusing previously learned visual representations could improve classification performance on the GTSRB dataset.

D. Training Strategy

All experiments used the same training configuration to ensure a fair comparison between augmentation techniques.

The training pipeline included:

- Adam optimizer;
- Cross-entropy loss;
- Early stopping;
- Learning-rate scheduling using ReduceLROnPlateau;

V. PHASE 1 - DATA AUGMENTATION STRATEGIES

The first phase is focused on evaluating the impact of data augmentation techniques on model performance.

Data augmentation is a key technique to improve the generalization ability of deep learning models, especially in image classification tasks such as traffic sign recognition. In this work, multiple augmentation strategies were systematically designed and evaluated using the GTSRB dataset.

The main objective was to analyze how different transformations affect model robustness and accuracy under real-world variations such as rotation, illumination changes, occlusions, and perspective distortion.

All experiments follow a controlled evaluation protocol using the same CNN architecture, optimizer (Adam), and training configuration, varying only the augmentation pipeline.

A. Experiment 0 - Baseline

The baseline experiment uses only minimal preprocessing, without any augmentation applied to the training data. This serves as the reference for all subsequent comparisons.

```
t_base = transform_base_vgg if model_name in ('vgg16',
'vgg16_tl') else transform_base
_, mean, std = run_experiment(
    name='baseline',
    transform_train=t_base,
    transform_test=t_base,
    model_name=model_name
)
```

B. Experiment 1 - Geometric Augmentation

This experiment introduces basic geometric transformations, aiming to improve invariance to small spatial changes such as rotation and translation.

```
transform_geo = v2.Compose([
    v2.RandomRotation(15, interpolation=v2.
InterpolationMode.BILINEAR),
    v2.Resize((IMG_SIZE_CONV, IMG_SIZE_CONV)),
    v2.RandomAffine(degrees=0, translate=(0.1, 0.1)),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True)
])

transform_geo_vgg = v2.Compose([
    v2.RandomRotation(15, interpolation=v2.
InterpolationMode.BILINEAR),
    v2.Resize((IMG_SIZE_VGG, IMG_SIZE_VGG)),
    v2.RandomAffine(degrees=0, translate=(0.1, 0.1)),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True),
    v2.Normalize([0.485, 0.456, 0.406],
[0.229, 0.224, 0.225])
])
```

These transformations simulate small camera shifts and orientation changes commonly found in real driving scenarios.

C. Experiment 2 - Advanced Geometric Augmentation

A more aggressive geometric pipeline was introduced using perspective distortion and random cropping. This aims to simulate more extreme viewpoint variations.

```
transform_geo2 = v2.Compose([
    v2.RandomPerspective(distortion_scale=0.2, p=0.6),
    v2.Resize((IMG_SIZE_CONV + 8, IMG_SIZE_CONV + 8)),
    v2.RandomCrop((IMG_SIZE_CONV, IMG_SIZE_CONV)),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True)
])

transform_geo2_vgg = v2.Compose([
    v2.RandomPerspective(distortion_scale=0.2, p=0.6),
    v2.Resize((IMG_SIZE_VGG + 8, IMG_SIZE_VGG + 8)),
    v2.RandomCrop((IMG_SIZE_VGG, IMG_SIZE_VGG)),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True),
    v2.Normalize([0.485, 0.456, 0.406],
                 [0.229, 0.224, 0.225])
])
```

This setup introduces stronger spatial variability while maintaining the semantic structure of traffic signs.

D. Experiment 3 - Color Augmentation

This experiment focuses on photometric transformations using ColorJitter operations. Brightness, contrast, saturation and hue were randomly modified to simulate different illumination and weather conditions.

```
transform_color = v2.Compose([
    v2.Resize((IMG_SIZE_CONV, IMG_SIZE_CONV)),
    v2.ColorJitter(brightness=0.5, contrast=0.5,
                  saturation=0.3, hue=0.05),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True)
])

transform_color_vgg = v2.Compose([
    v2.Resize((IMG_SIZE_VGG, IMG_SIZE_VGG)),
    v2.ColorJitter(brightness=0.5, contrast=0.5,
                  saturation=0.3, hue=0.05),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True),
    v2.Normalize([0.485, 0.456, 0.406],
                 [0.229, 0.224, 0.225])
])
```

Color jittering is particularly relevant for real-world driving scenarios where illumination is highly variable.

E. Experiment 4 - Combined Augmentation

This experiment combines geometric and color augmentations, along with random erasing to simulate occlusions such as dirt, shadows, or partial obstruction.

```
transform_combined = v2.Compose([
    v2.RandomRotation(15, interpolation=v2.
InterpolationMode.BILINEAR),
    v2.Resize((IMG_SIZE_CONV, IMG_SIZE_CONV)),
    v2.RandomAffine(degrees=0, translate=(0.1, 0.1)),
    v2.ColorJitter(brightness=0.5, contrast=0.5,
                  saturation=0.3, hue=0.05),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True),
```

```
    v2.RandomErasing(p=0.3, scale=(0.02, 0.1),
                    value='random')
])

transform_combined_vgg = v2.Compose([
    v2.RandomRotation(15, interpolation=v2.
InterpolationMode.BILINEAR),
    v2.Resize((IMG_SIZE_VGG, IMG_SIZE_VGG)),
    v2.RandomAffine(degrees=0, translate=(0.1, 0.1)),
    v2.ColorJitter(brightness=0.5, contrast=0.5,
                  saturation=0.3, hue=0.05),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True),
    v2.RandomErasing(p=0.3, scale=(0.02, 0.1),
                    value='random'),
    v2.Normalize([0.485, 0.456, 0.406],
                 [0.229, 0.224, 0.225])
])
```

Random erasing improves robustness by forcing the network to learn partial-feature representations.

F. Experiment 5 - Massive Dataset Expansion

This experiment significantly increases dataset diversity by duplicating the training dataset with a strongly augmented version. The final dataset is a concatenation of original and augmented samples.

```
transform_strong = v2.Compose([
    v2.RandomRotation(20, interpolation=v2.
InterpolationMode.BILINEAR),
    v2.Resize((IMG_SIZE_CONV, IMG_SIZE_CONV)),
    v2.RandomAffine(degrees=0, translate=(0.1, 0.1)),
    v2.ColorJitter(brightness=0.5, contrast=0.5,
                  saturation=0.3, hue=0.05),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True),
    v2.RandomErasing(p=0.3, scale=(0.02, 0.1),
                    value='random')
])

transform_strong_vgg = v2.Compose([
    v2.RandomRotation(20, interpolation=v2.
InterpolationMode.BILINEAR),
    v2.Resize((IMG_SIZE_VGG, IMG_SIZE_VGG)),
    v2.RandomAffine(degrees=0, translate=(0.1, 0.1)),
    v2.ColorJitter(brightness=0.5, contrast=0.5,
                  saturation=0.3, hue=0.05),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True),
    v2.RandomErasing(p=0.3, scale=(0.02, 0.1),
                    value='random'),
    v2.Normalize([0.485, 0.456, 0.406],
                 [0.229, 0.224, 0.225])
])
```

This strategy effectively doubles the dataset size, improving generalization capacity.

The model is trained multiple times with different random initializations to ensure statistical reliability:

```
for i in range(RUNS):
    if model_name == 'conv_II':
        model = Conv_II(NUM_CLASSES).to(device)
    elif model_name == 'vgg16':
        model = torchvision.models.vgg16()
    ...
    train(...)
acc = evaluate(model, test_loader)
```

This setup also includes:

- Learning rate scheduling (ReduceLROnPlateau);
- Early stopping;
- Model checkpointing.

G. Experiment 6 - MixUp and CutMix Augmentation

This experiment evaluates the impact of advanced data augmentation techniques, specifically MixUp and CutMix, on model generalization. Both methods generate synthetic training samples to improve robustness and reduce overfitting.

MixUp creates new training examples by linearly interpolating pairs of images and their corresponding labels, producing soft labels that encourage smoother decision boundaries. Meanwhile, CutMix improves augmentation diversity by replacing a random patch of one image with a patch from another image, while adjusting the labels proportionally to the area of the patch.

```

mixup = v2.MixUp(num_classes=NUM_CLASSES)
history = {'loss': [], 'accuracy': [],
          'val_loss': [], 'val_acc': []}
best_val_loss = np.inf

...

transform_mix = v2.Compose([
    v2.Resize((IMG_SIZE_CONV, IMG_SIZE_CONV)),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True),
])

transform_mix_vgg = v2.Compose([
    v2.Resize((IMG_SIZE_VGG, IMG_SIZE_VGG)),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True),
    v2.Normalize([0.485, 0.456, 0.406],
                 [0.229, 0.224, 0.225])
])

```

Unlike standard training, MixUp produces soft labels, improving decision boundaries and reducing overfitting. CutMix (when applied) similarly acts as a regularization technique by enforcing the model to learn from partial image regions and mixed labels.

During evaluation, standard validation is performed without MixUp or CutMix to ensure fair performance measurement.

VI. PHASE 1 RESULTS AND DISCUSSION

Accuracy Scores			
Model \ Data Augmentation	ConvII	VGG16	VGG16_tl
Baseline	97.28%	5.7%	86.66%
Geo	96.60%	5.7%	88.07%
Geo2	97.43%	5.7%	87.40%
Color	97.13%	5.7%	86.51%
Combined	97.59%	5.7%	86.14%
Mix	97.76%	5.7%	86.61%
Massive Expansion	98.57%	5.7%	89.84%

The experimental results demonstrate that data augmentation has a significant influence on classification performance.

Several observations can be highlighted:

- Baseline models already achieved strong performance due to the relatively large size and quality of the GTSRB dataset.
- Moderate geometric transformations improved robustness to viewpoint changes, although excessively aggressive transformations occasionally degraded performance by producing unrealistic samples.
- Photometric augmentation improved robustness to illumination variations while maintaining stable accuracy.
- Combined augmentation increased training diversity but also introduced additional optimization difficulty.
- Massive dataset expansion consistently produced some of the strongest results, indicating that increasing sample diversity is highly beneficial for traffic sign recognition.
- MixUp and CutMix regularization improved generalization by encouraging smoother class boundaries and reducing overfitting.

In the image below, we can observe the confusion matrix for the Massive Expansion experiment for the VGG16_tl model.

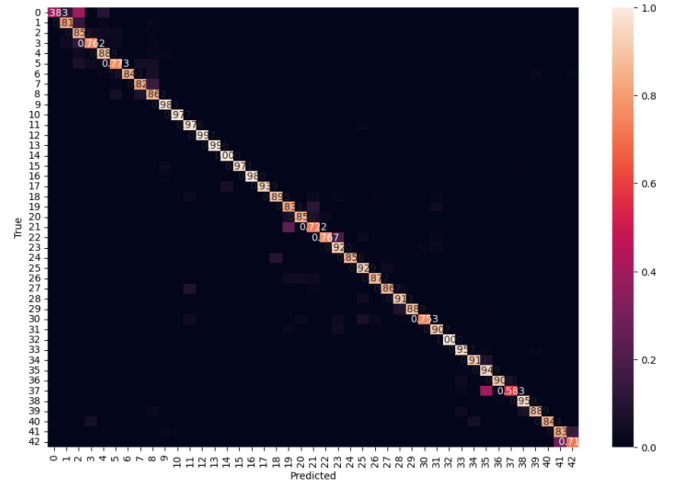


Figure 1. Confusion Matrix - VGG16_tl, Massive Expansion

The confusion matrix confirms the strong performance of the model, as most predictions are concentrated along the main diagonal. This indicates that the majority of traffic sign classes are correctly classified, with high recall values across most categories.

The most significant misclassification occurs for class 0 (Speed Limit 20 km/h), which achieves a diagonal value of 0.383. A considerable proportion of these samples are incorrectly classified as class 2 (Speed Limit 50 km/h). This confusion is likely caused by the strong visual similarity between speed limit signs, which share the same circular shape, color scheme, and overall design, differing primarily in the numerical value displayed.

Another notable source of error is observed for class 37 (Go Straight or Turn Left), whose diagonal value is 0.583. The model frequently confuses this class with class 35 (Go Straight). These signs contain similar directional arrow patterns and visual characteristics, making them more difficult to distinguish,

particularly under challenging imaging conditions such as low resolution, occlusion, or varying illumination.

Despite these specific cases, the confusion matrix demonstrates that the model is able to discriminate effectively between most traffic sign categories, supporting the high overall accuracy achieved in the Massive Expansion experiment.

VII. PHASE 2 – ENSEMBLE LEARNING

Although individual neural networks achieved strong performance, different models tend to make different classification errors. Ensemble learning exploits this diversity by combining multiple predictors into a single decision system.

The objective of Phase 2 was therefore to evaluate whether combining models trained during Phase 1 could further improve classification accuracy.

A. Ensemble Construction

In the second phase of the project, ensemble learning techniques were explored to evaluate whether combining multiple trained models could improve classification performance.

For each augmentation strategy, several models obtained from independent training runs were grouped into an ensemble. Since each model may learn slightly different decision boundaries due to random initialization and training dynamics, combining their predictions can reduce variance and improve robustness.

Two ensemble approaches were evaluated: Soft Voting and Hard Voting.

B. Soft Voting

In the Soft Voting approach, the output scores produced by each model are combined, and the final prediction is determined by the class receiving the highest overall score. This method takes into account the confidence of each model and is often more effective when the individual classifiers provide reliable probability estimates.

C. Hard Voting

In the Hard Voting approach, each model independently predicts a class label. The final prediction is then obtained through majority voting, where the class receiving the largest number of votes is selected.

Unlike Soft Voting, this method only considers the final decision of each model and ignores the confidence associated with the predictions.

D. Experimental Evaluation

Ensembles were evaluated using different combinations of independently trained CNN models and transfer learning variants (VGG16-based). Each ensemble combines multiple checkpoints trained under different experimental configurations (e.g., baseline and augmentation-based variants).

The evaluated ensemble configurations are:

- **ConvII + VGG16 (baseline):** combination of baseline ConvII and VGG16 transfer learning models;

- **ConvII multi-augmentation ensemble:** includes baseline, geometric transformations, advanced geometric augmentation, color jittering, combined augmentations, mix-based strategies and expansion-based training;
- **VGG16 multi-augmentation ensemble:** same augmentation family applied to the VGG16 transfer learning model;
- **Best-performing augmentation ensemble:** combines ConvII and VGG16 models trained with expansion and selected high-performing augmentation strategies;
- **Worst-case augmentation ensemble:** combines ConvII and VGG16 models trained with lower-performing configurations (e.g., geometric and combined variants).

For each ensemble, we evaluate performance using both soft voting and hard voting strategies:

- Soft Voting accuracy, obtained by averaging logits across models before applying the argmax decision;
- Hard Voting accuracy, obtained by majority voting over individual model predictions.

VIII. PHASE 2 RESULTS AND DISCUSSION

Accuracy Scores - Ensembles		
Ensemble	Soft Voting	Hard Voting
Baseline - VGG16_tl and ConvII	96.33%	97.28%
All experiments on ConvII	98.57%	98.49%
All experiments on VGG16_tl	91.11%	91.01%
Best overall experiments	97.61%	98.80%
Worst overall experiments	96.53%	96.60%

Ensemble learning provides two important advantages.

First, averaging predictions reduces variance and mitigates errors caused by individual model initialization.

Second, combining models trained under different random conditions improves robustness and typically increases generalization performance.

The experimental results show that ensemble methods consistently improve performance over individual models, with the best configuration achieving up to **98.80%** accuracy using hard voting. Soft voting generally provides smoother aggregation, while hard voting occasionally yields slightly higher accuracy depending on model diversity and complementarity.

IX. CONCLUSION

This project explored the application of deep learning techniques to traffic sign recognition using the GTSRB dataset.

In the first phase, several data augmentation strategies were evaluated, including geometric transformations, color augmentation, dataset expansion, and MixUp. The results showed that data augmentation plays an important role in improving model generalization and robustness, with stronger augmentation pipelines generally producing better performance.

In the second phase, ensemble learning techniques were investigated by combining models trained during Phase 1. Both Hard Voting and Soft Voting approaches were evaluated, demonstrating that ensembles can further improve classification performance by leveraging the diversity of individual models.

In most configurations, soft voting provided more stable improvements by aggregating class probabilities, while hard voting occasionally achieved slightly higher accuracy when model predictions were highly consistent.

The experimental results confirm that ensembles consistently outperform individual models, with the best configuration achieving up to 98.80% accuracy. This highlights that model diversity, introduced through different augmentation strategies and training setups, is the main factor in improving ensemble effectiveness.

Overall, the experiments confirm that combining effective data augmentation with ensemble learning contributes to a significant improvement in traffic sign classification performance and robustness.

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